

Course

« *Deep Learning* »

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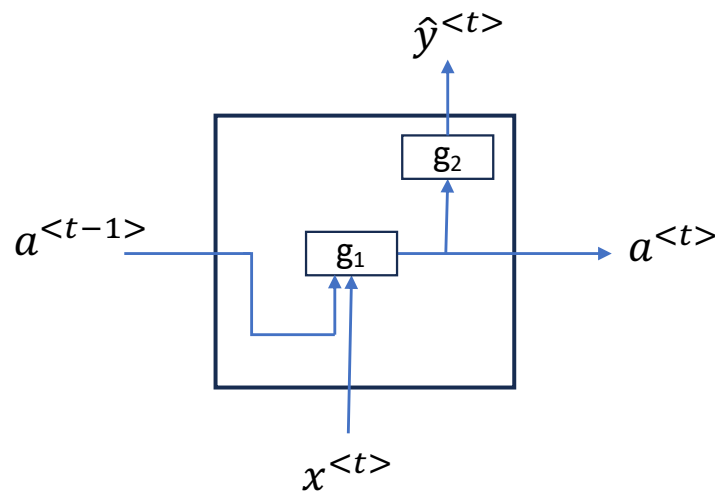
Sequence models: Gated Recurrent Unit (GRU)

- Principle

Improving RNN unit to:

- Cope better with long range context and
- Better address the vanishing gradient problem.

- Let's recall the RNN unit



$$a^{<t>} = g_1(w_a \cdot [a^{<t-1>}, x^{<t>}] + b_a)$$

$$\hat{y}^{<t>} = g_2(w_y \cdot a^{<t>} + b_y)$$

Sequence models: Gated Recurrent Unit (GRU)

- A simplified version of GRU
 - Introduction of the memory cell c .
 - c will provide the cell with a certain memory.
 - c and a refer to the same entity ($c^{<t>} = a^{<t>}$), but to make a difference between RNN and GRU, we will use c .

- The main equation of a simplified GRU is:

$$\tilde{c}^{<t>} = \tanh(w_c \cdot [c^{<t-1>}, x^{<t>}] + b_c)$$

$\tilde{c}^{<t>}$ is a candidate for the new value of $c^{<t>}$

- And we introduce an **update gate** (Γ_u):

$$\Gamma_u^{<t>} = \sigma(w_u \cdot [c^{<t-1>}, x^{<t>}] + b_u)$$

Here σ is the sigmoid $\Rightarrow$ The output of $\Gamma_u^{<t>} \in [0, 1]$

- $\Gamma_u^{<t>}$ governs the update of $c^{<t>}$ (using $\tilde{c}^{<t>}$ and $c^{<t-1>}$): $c^{<t>} = \Gamma_u^{<t>} * \tilde{c}^{<t>} + (1 - \Gamma_u^{<t>}) * c^{<t-1>}$

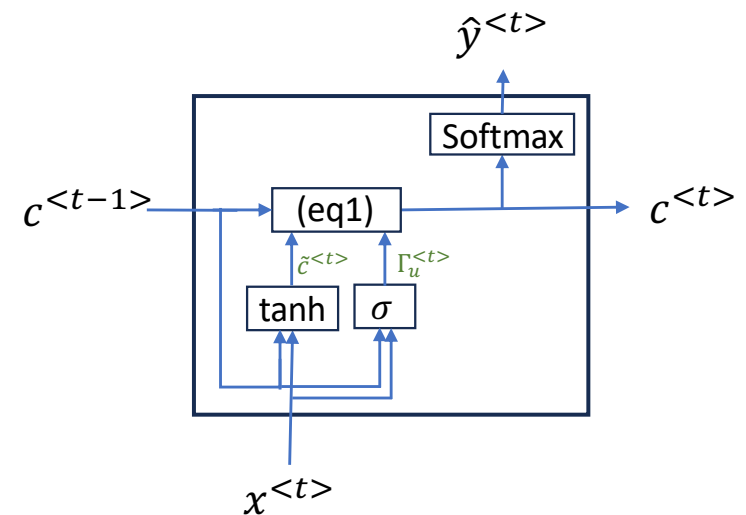
Sequence models: Gated Recurrent Unit (GRU)

A simplified version of GRU

- $\tilde{c}^{<t>} = \tanh(w_c \cdot [c^{<t-1>}, x^{<t>}] + b_c)$
- $\Gamma_u^{<t>} = \sigma(w_u \cdot [c^{<t-1>}, x^{<t>}] + b_u)$
- $c^{<t>} = \Gamma_u^{<t>} * \tilde{c}^{<t>} + (1 - \Gamma_u^{<t>}) * c^{<t-1>} \quad (\text{eq1})$

Very important: * in eq1 is element-wise.

These element-wise multiplications just tells you GRU which are the dimensions of the memory cell vector to update at every time step.



$\Gamma_u = 1$ 0 0 0 ...

↓ ↓ ↓ ↓

0 $\Gamma_u = 1$

↓ ↓

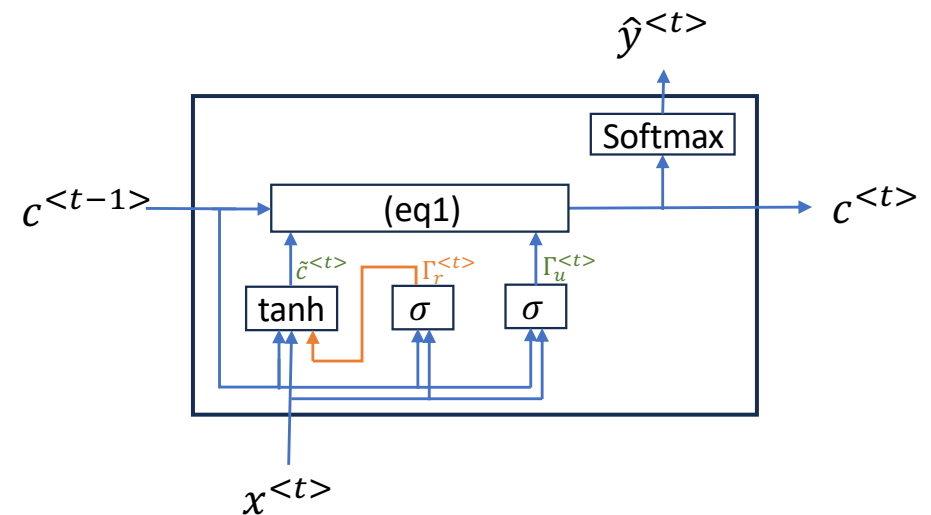
“The professor announced that the papers submitted by the student, which had been reviewed thoroughly by the committee, were now ready to be published.”

One bit from Γ_u can be used to remember the singular or plural of the word « paper ».
Other bits can be used to tell that we are talking about studies.

Sequence models: Gated Recurrent Unit (GRU)

A full version of GRU

- The introduction of a **relevance gate** (Γ_u).
- $\tilde{c}^{<t>} = \tanh(w_c \cdot [\Gamma_r^{<t>} * c^{<t-1>}, x^{<t>}] + b_c)$
- $\Gamma_r^{<t>} = \sigma(w_r \cdot [c^{<t-1>}, x^{<t>}] + b_r)$
- $\Gamma_u^{<t>} = \sigma(w_u \cdot [c^{<t-1>}, x^{<t>}] + b_u)$
- $c^{<t>} = \Gamma_u^{<t>} * \tilde{c}^{<t>} + (1 - \Gamma_u^{<t>}) * c^{<t-1>} \quad (eq1)$



In the literature:

- Different notations.
- Many different variants of GRU units.

Sequence models: Long Short Term Memory (LSTM)

- In LSTM:
 - More gates are introduced (w.r.t. GRU).
 - $c^{<t>}$ and $a^{<t>}$ are used together and they encode different information.

GRU

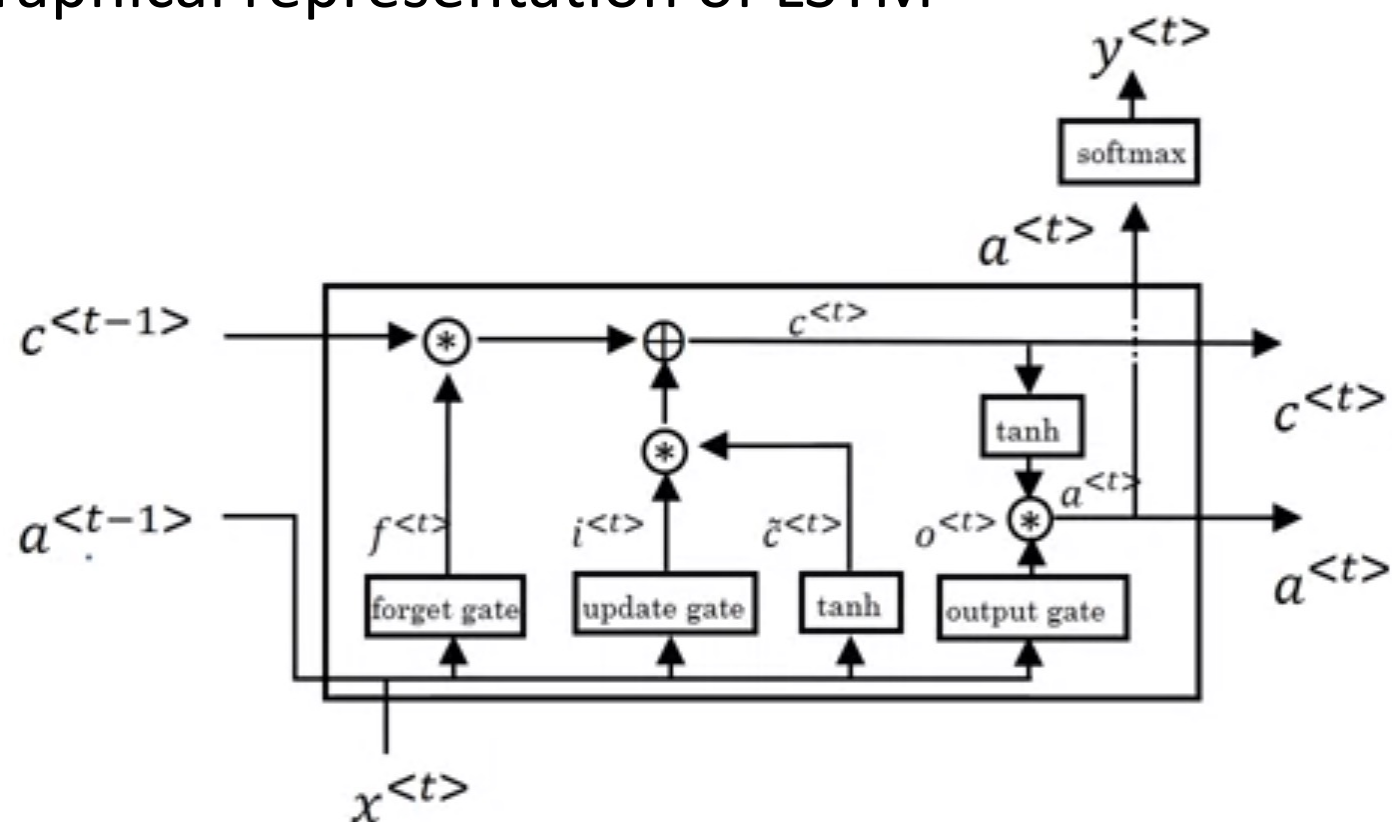
- $\tilde{c}^{<t>} = \tanh(w_c \cdot [\Gamma_r^{<t>} * c^{<t-1>}, x^{<t>}] + b_c)$
- $\Gamma_r^{<t>} = \sigma(w_r \cdot [c^{<t-1>}, x^{<t>}] + b_r)$
- $\Gamma_u^{<t>} = \sigma(w_u \cdot [c^{<t-1>}, x^{<t>}] + b_u)$
- $c^{<t>} = \Gamma_u^{<t>} * \tilde{c}^{<t>} + (1 - \Gamma_u^{<t>}) * c^{<t-1>}$
- $a^{<t>} = c^{<t>}$

LSTM

- $\tilde{c}^{<t>} = \tanh(w_c \cdot [a^{<t-1>}, x^{<t>}] + b_c)$
- $\Gamma_u^{<t>} = \sigma(w_u \cdot [a^{<t-1>}, x^{<t>}] + b_u)$ – Update
- $\Gamma_f^{<t>} = \sigma(w_f \cdot [a^{<t-1>}, x^{<t>}] + b_f)$ – Forget
- $\Gamma_o^{<t>} = \sigma(w_o \cdot [a^{<t-1>}, x^{<t>}] + b_o)$ – Output
- $c^{<t>} = \Gamma_u^{<t>} * \tilde{c}^{<t>} + \Gamma_f^{<t>} * c^{<t-1>}$
- $a^{<t>} = \Gamma_o^{<t>} * \tanh(c^{<t>})$

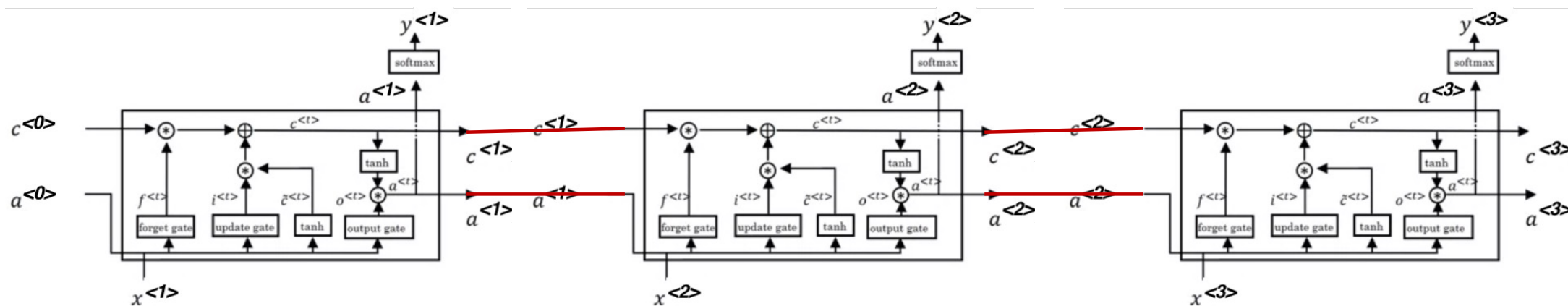
Sequence models: Long Short Term Memory (LSTM)

- Graphical representation of LSTM



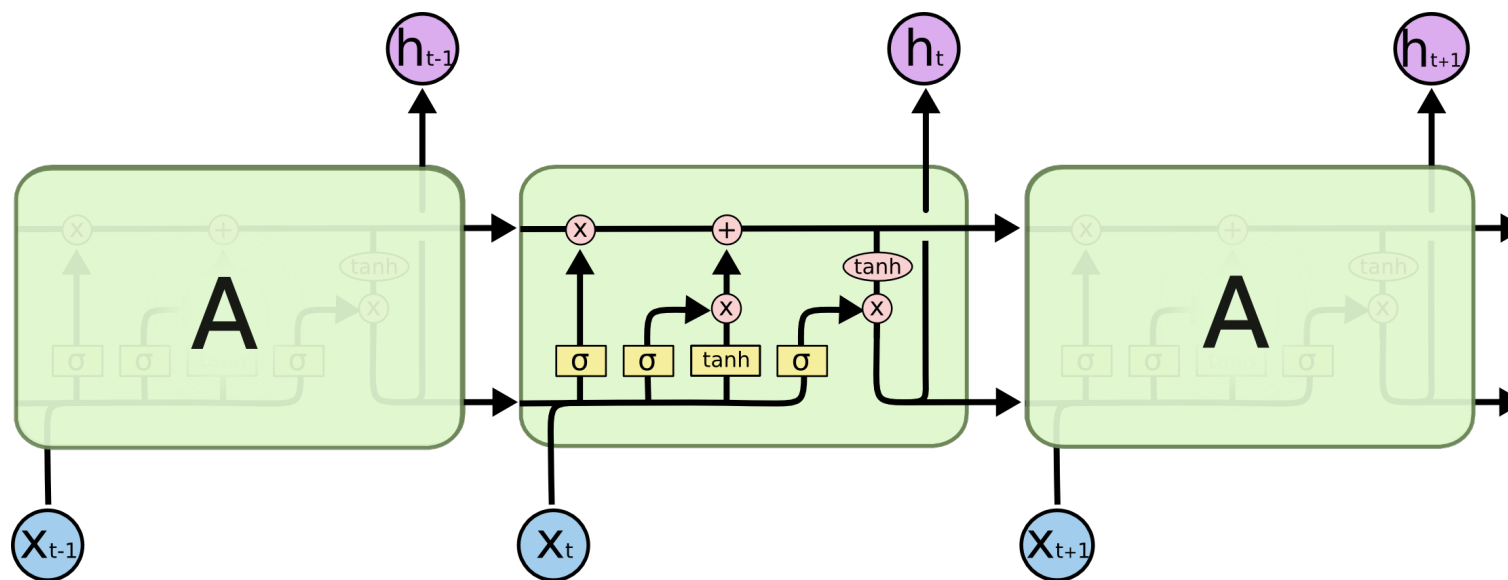
Sequence models: Long Short Term Memory (LSTM)

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Sequence models: Long Short Term Memory (LSTM)

- Graphical representation of LSTM



<https://colah.github.io/posts/2015-08-Understanding-LSTMs/>

Sequence models: Long Short Term Memory (LSTM)

A variant of LSTM: Using peephole connections

- Peephole connections are a variation of the traditional LSTM architecture
- They have been introduced to improve its ability to model precise temporal dependencies.
- In standard LSTM, the cell state $c^{<t>}$ is not directly involved in computing the values of the gates (input, forget, and output gates).
- Peephole connections address this limitation by allowing the cell state to directly influence the gate computations.

Sequence models: Long Short Term Memory (LSTM)

A variant of LSTM: Using peephole connections

- $\tilde{c}^{<t>} = \tanh(w_c \cdot [a^{<t-1>}, x^{<t>}] + b_c)$
- $\Gamma_u^{<t>} = \sigma(w_u \cdot [a^{<t-1>}, x^{<t>}, c^{<t-1>}] + b_u)$ – **Update**
- $\Gamma_f^{<t>} = \sigma(w_f \cdot [a^{<t-1>}, x^{<t>}, c^{<t-1>}] + b_f)$ – **Forget**
- $\Gamma_o^{<t>} = \sigma(w_o \cdot [a^{<t-1>}, x^{<t>}, c^{<t-1>}] + b_o)$ – **Output**
- $c^{<t>} = \Gamma_u^{<t>} * \tilde{c}^{<t>} + \Gamma_f^{<t>} * c^{<t-1>}$
- $a^{<t>} = \Gamma_o^{<t>} * \tanh(c^{<t>})$

Sequence models: Long Short Term Memory (LSTM)

- LSTMs actually came much earlier than GRUs
- GRUs are relatively recent
- GRUs can be seen as a simplification of the more complicated LSTM model.

Sequence models: Making use also of the future

- Bidirectional RNN

RNN seen so far are unidirectional: At each time step, the input is processed taking into account only past inputs.

In practice, this is not sufficient in the general case!

Examples :

1. She gave her friend a **ring**, but he didn't answer.

ring = a piece of jewelry

ring = a phone call

2. The **bank** was crowded with people, some waiting in line to deposit money and others enjoying the cool shade of the trees.

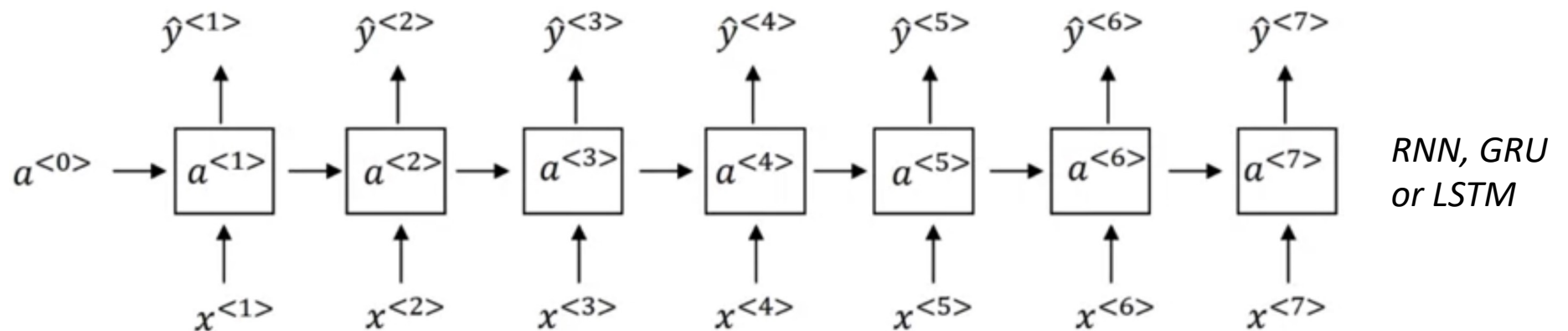
bank = a financial institution

bank = the side of a river

Sequence models: Bidirectional RNN

- Bidirectional RNN

RNN seen so far are unidirectional

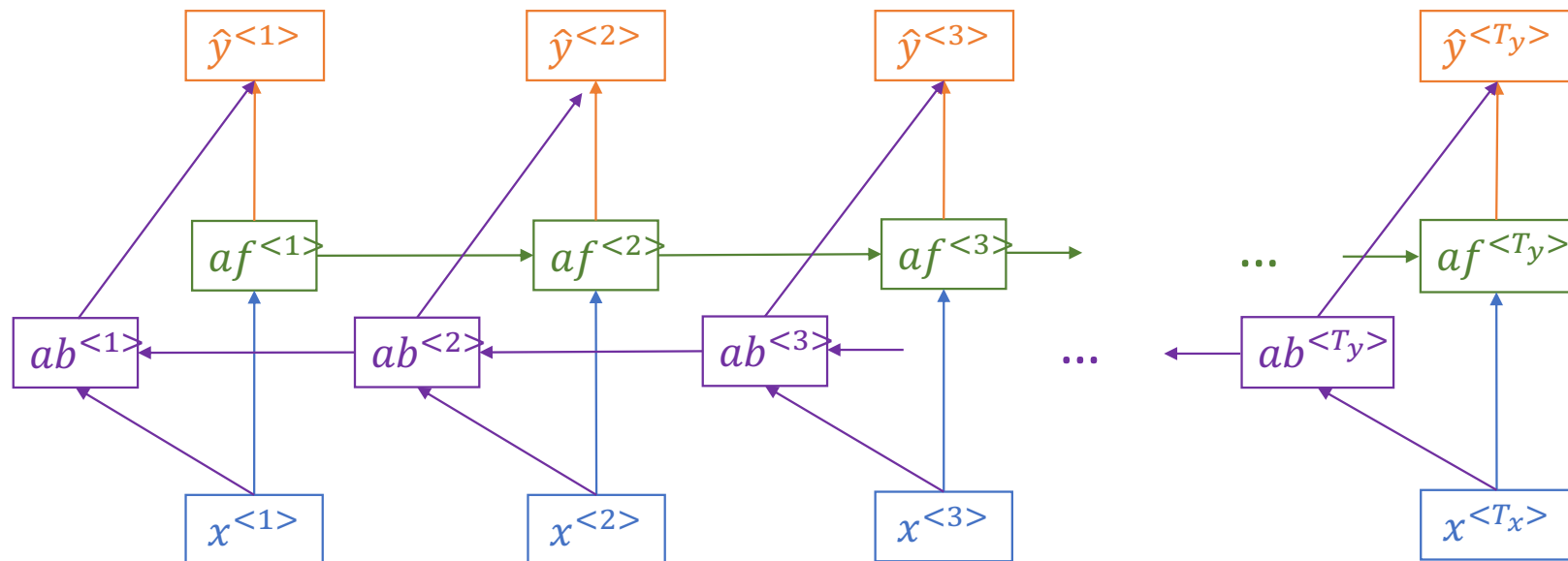


Sequence models: Bidirectional RNN

- Bidirectional RNN

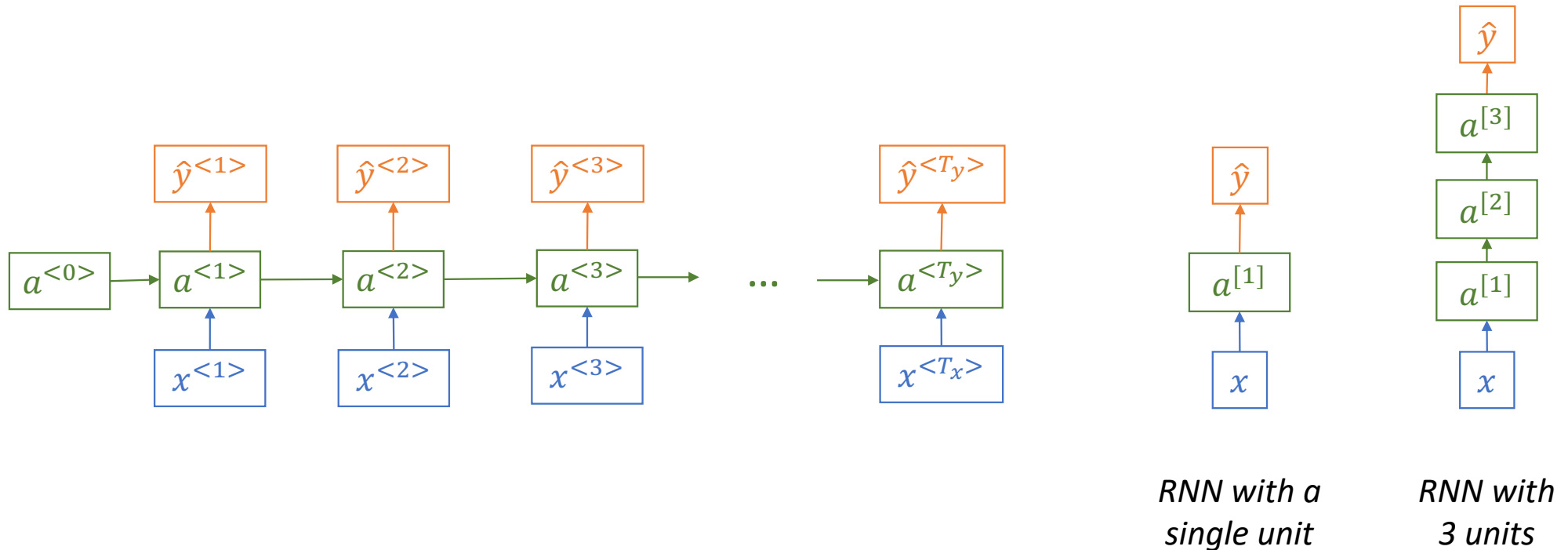
RNN seen so far are unidirectional

$$\hat{y}^{<t>} = g(w_y \cdot [af^{<t>}, ab^{<t>}] + b_y)$$



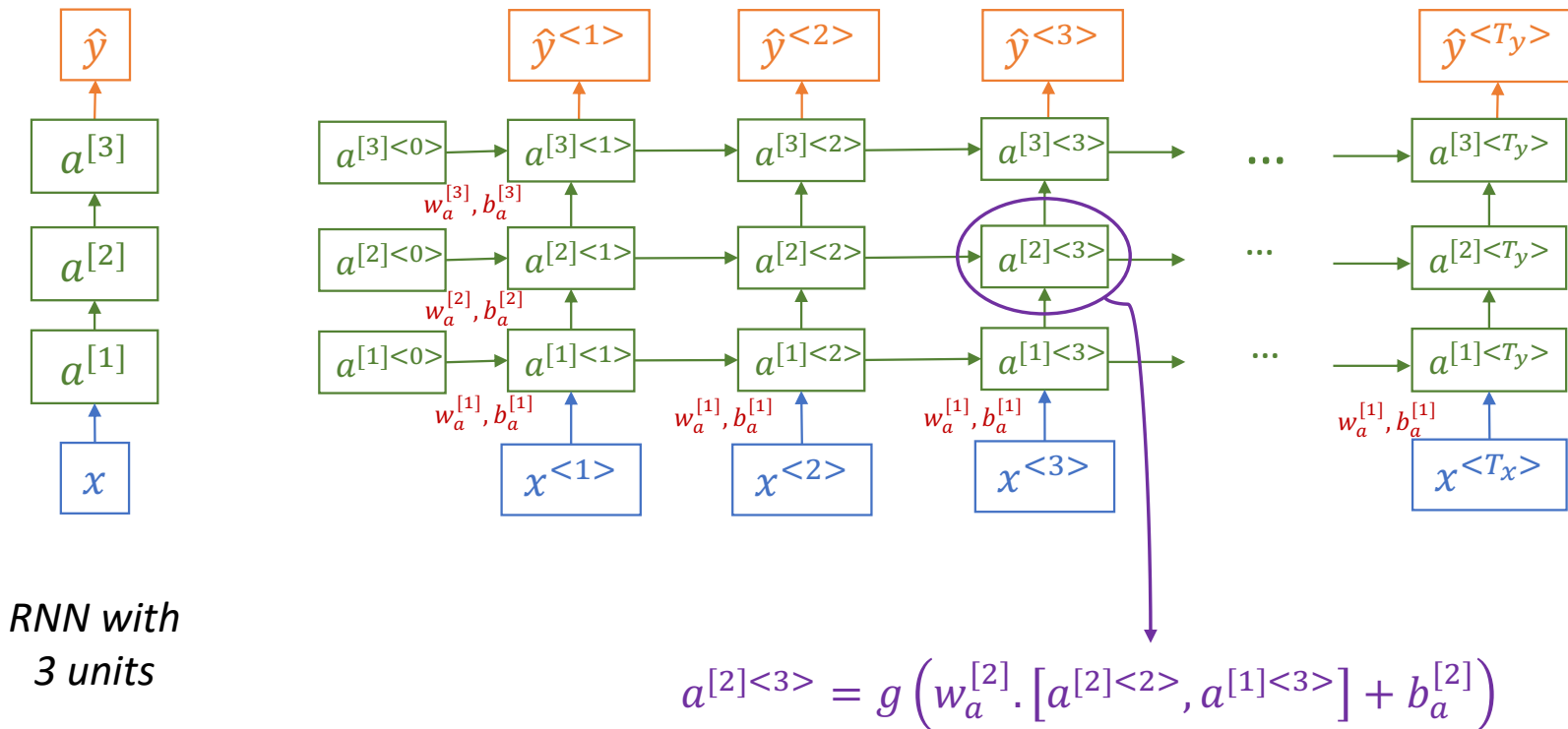
Sequence models: Deep RNN

- Introducing more depth into RNN (Simple RNN, GRU or LSTM)



Sequence models: Deep RNN

- Introducing depth into RNN (Simple RNN, GRU or LSTM)



RNN with
3 units